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Taller 2
Sistemas Dinámicos y Control
Lazo cerrado y Lugar Geométrico de las Raíces

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Problemas del libro "Feedback Control of Dynamic Systems – Eight Edition – Franklin, Powell, Emami-Naeini, Pearson".

Capítulo 4: Sistemas en lazo cerrado y PID

Control del error en estado estacionario

1. Problema 4.6 – página 252
2. Problema 4.8 – página 253
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4. Problema 4.29 – página 261

Control PID

5. The transfer functions of speed control for a magnetic tape-drive system are shown in Fig. 1. The speed sensor is fast enough that its dynamics can be neglected and the diagram shows the equivalent unity feedback system.

    a) Assuming the reference is zero, what is the steady-state error due to a step disturbance torque of 1 Nm? What must the amplifier gain K be in order to make the steady-state error ess < 0.001 rad/sec.?
    b) Plot the roots of the closed-loop system in the complex plane, and accurately sketch the time response of the output for a step reference input using the gain K computed in part (a).
    c) Plot the region in the complex plane of acceptable closed-loop poles corresponding to the specifications of a 1% settling time of ts<0.1 sec. and an overshoot Mp < 5%.
    d) Give values for kp and kD for a PD controller which will meet the specifications.
    e) How would the disturbance-induced steady-state error change with the new control scheme in part (d)? How could the steady-state error to a disturbance torque be eliminated entirely

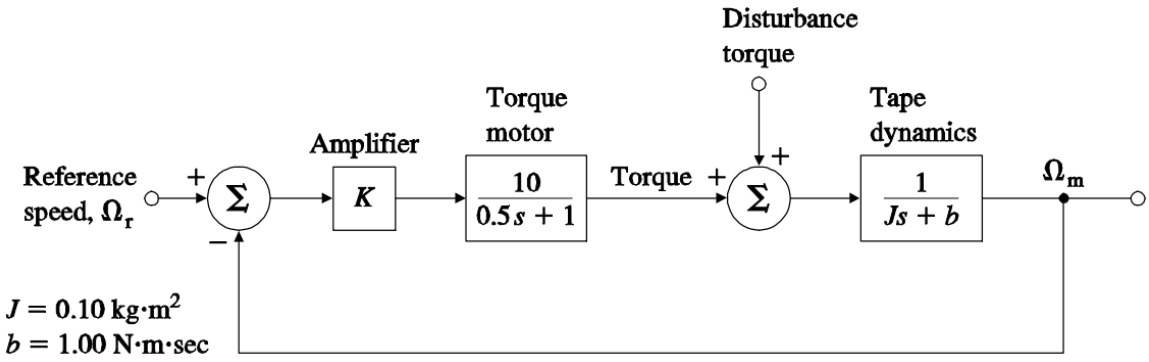


Fig. 1

6. The DC-motor speed control shown in Fig. 4.41 is described by the differential equation

$$\dot{y} + 60y = 600v_a - 1500w$$

where y is the motor speed, va is the armature voltage, and w is the load torque. Assume the armature voltage is computed using the PI control law

$$v_a = -\left(k_p e + k_I \int_0^t e dt\right)$$

$$e = r - y.$$

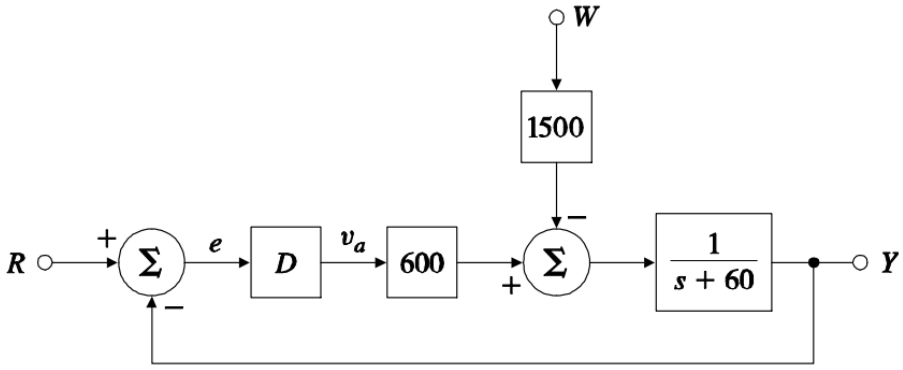


Fig. 2. D.C. Motor speed control block diagram

a) Compute the transfer function from W to Y as a function of kp and kI.
b) Compute values for kp and kI so that the characteristic equation of the closed-loop system will have roots at 60 ± 60j.

Capítulo 5: Lugar Geométrico de las Raíces

Lugar geométrico de un sistema en lazo cerrado básico

7. Problema 5.1 – página 335

Lineamientos para bosquejar el lugar geométrico de las raíces

8. Problema 5.7 – página 338
9. Problema 5.10 – página 339

Ejemplos de diseño de compensadores utilizando el lugar geométrico de las raíces

10. Problema 5.21 – página 341
11. Problema 5.24 – página 342
12. Problema 5.31 – página 344

4.3 Compare the two structures shown in Fig. 4.31 with respect to sensitivity to changes in the overall gain due to changes in the amplifier gain. Use the relation

$$S = \frac{d \ln F}{d \ln K} = \frac{K}{F}\frac{dF}{dK},$$

as the measure. Select H_1 and H_2 so the nominal system outputs satisfy $F_1 = F_2$, and assume $KH_1 > 0$.

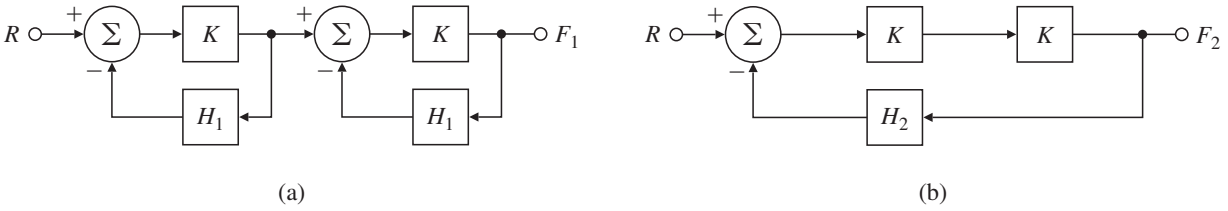


Figure 4.31
Block diagrams for Problem 4.3

4.4 A unity feedback control system has the open-loop transfer function

$$G(s) = \frac{A}{s(s+a)}.$$

(a) Compute the sensitivity of the closed-loop transfer function to changes in the parameter A.

(b) Compute the sensitivity of the closed-loop transfer function to changes in the parameter a.

(c) If the unity gain in the feedback changes to a value of $\beta \neq 1$, compute the sensitivity of the closed-loop transfer function with respect to β.

4.5 Compute the equation for the system error for the feedback system shown in Fig. 4.5.

Problems for Section 4.2: Control of Steady-State Error

4.6 Consider the DC motor control system with rate (tachometer) feedback shown in Fig. 4.32(a).

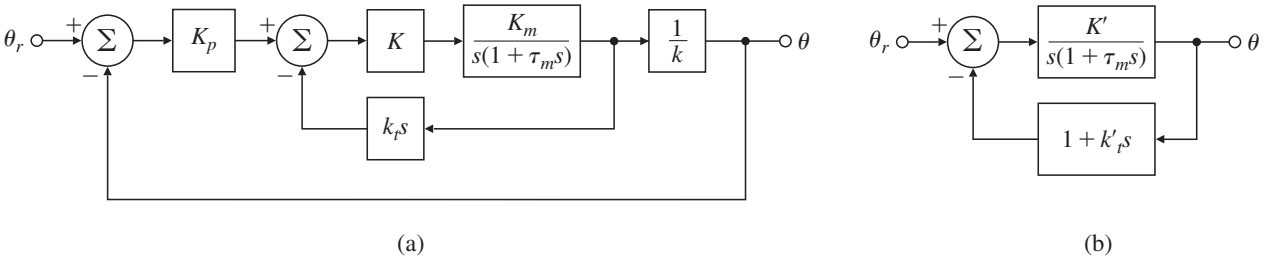


Figure 4.32
Control system for Problem 4.6

(a) Find values for K' and k'_t so the system of Fig. 4.32(b) has the same transfer function as the system of Fig. 4.32(a).

(b) Determine the system type with respect to tracking θ_r and compute the system K_v in terms of parameters K' and k'_t.

(c) Does the addition of tachometer feedback with positive k_t increase or decrease K_v?

4.7 A block diagram of a control system is shown in Fig. 4.33.

(a) If r is a step function and the system is closed-loop stable, what is the steady-state tracking error?

(b) What is the system type?

(c) What is the steady-state error to a ramp velocity 2.5 if $K_2 = 2$ and K_1 is adjusted so that the system step response approximately has a rise time of 0.65 s and a settling time of 0.23 s?

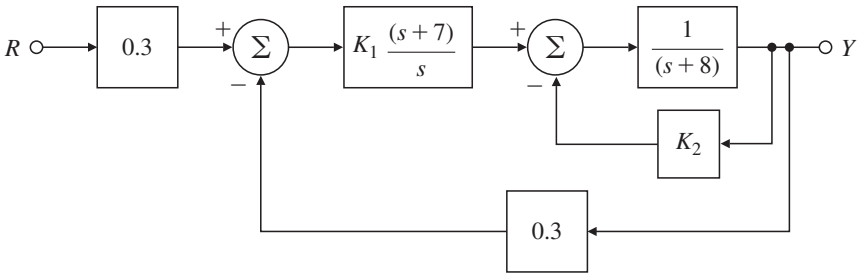


Figure 4.33
Closed-loop system for Problem 4.7

4.8 A standard feedback control block diagram is shown in Fig. 4.5 with

$$G(s) = \frac{1.5}{s};\ D_c(s) = \frac{(s+9)}{(s+3)};\ H(s) = \frac{70}{(s+70)};\ V_{(s)} = 0.$$

(a) Let $W = 0$ and compute the transfer function from R to Y.

(b) Let $R = 0$ and compute the transfer function from W to Y.

(c) What is the tracking error if R a unit-step input and $W = 0$?

(d) What is the tracking error if R is a unit-ramp input and $W = 0$?

(e) What is the system type with respect to the reference inputs and the corresponding error coefficient?

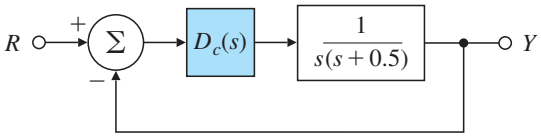


Figure 4.34
Control system for Problem 4.10

4.9 A generic negative feedback system with non-unity transfer function in the feedback path is shown in Fig. 4.5.

(a) Suppose,

$$G(s) = \frac{1}{s(s+1)^2};\ D_{cl}(s) = 0.42;\ H(s) = P\frac{(0.58s+1)}{(0.35s+1)};\ V(s) = 0,$$

showing a lead compensation in the feedback path. What is the requirement on P such that the system will remain a Type 1 system with respect to the reference input?

(b) For part (a), find the steady-state tracking error for this system to a unit ramp reference input if $P = 1$.

(c) For part (b), what is the value of the velocity error coefficient, K_v?

4.10 Consider the system shown in Fig. 4.34 where

$$D_c(s) = K\frac{(s^2 + \alpha s + 1)}{(s^2 + \omega_o^2)}.$$

(a) Prove that if the system is stable, it is capable of tracking a sinusoidal reference input $r = \sin \omega_o t$ with a zero steady-state error. (*Hint:* Look at the transfer function from R to E and consider the gain at ω_o.)

(b) Use the Routh's criterion to find the range of K such that the closed-loop system remains stable if $\omega_o = 1$ and $\alpha = 0.3$.

4.11 Consider the system shown in Fig. 4.35, which represents control of the angle of a pendulum that has no damping.

(a) What condition must $D_c(s)$ satisfy so the system can track a ramp reference input with constant steady-state error?

(b) For a transfer function $D_c(s)$ that stabilizes the system and satisfies the condition in part (a), find the class of disturbances $w(t)$ that the system can reject with zero steady-state error.

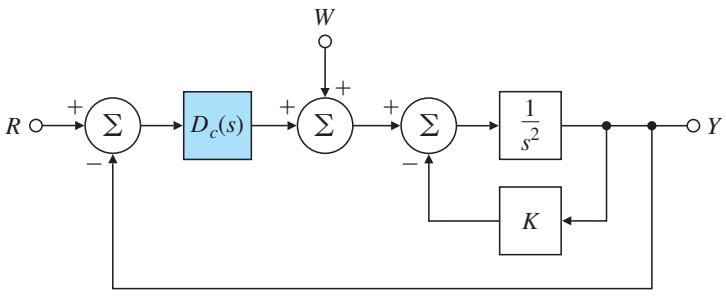


Figure 4.35
Control system for Problem 4.11

4.12 A unity feedback system has the overall transfer function

$$\frac{Y(s)}{R(s)} = \mathcal{T}(s) = \frac{6}{(7s + 2)(s + 3)}.$$

Give the system type and the corresponding error constant for tracking polynomial reference inputs in terms of ζ and ω_n.

4.13 Consider the second-order system

$$G(s) = \frac{2}{s^2 + 4\zeta s + 2}.$$

We would like to add a transfer function of the form $D_c(s) = K(s + a)/(s + b)$ in cascade with $G(s)$ in a unity-feedback structure.

(a) Ignoring stability for the moment, what are the constraints on K, a, and b so that the system is Type 1?

(b) What are the constraints on a, and b so that the system is both Type 1 and remains stable for every positive value for K?

$$\lim_{t\to\infty} y(t) = r_o.$$

Include both the open-loop (assuming $H_y = 0$) and feedback cases ($H_y \neq 0$) in your discussion.

(c) Repeat part (b) assuming a constant grade disturbance $W(s) = \frac{w_o}{s}$ is present *in addition to* the reference input. In particular, find the variation in speed due to the grade change for both the feedforward and feedback cases. Use your results to explain (1) why feedback control is necessary and (2) how the gain k_P should be chosen to reduce steady-state error.

(d) Assume $w(t) = 0$ and the gain A undergoes the perturbation $A + \delta A$. Determine the error in speed due to the gain change for both the feedforward and feedback cases. How should the gains be chosen in this case to reduce the effects of δA?

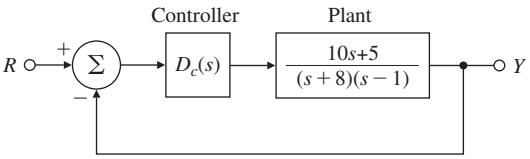
4.28 Prove that the step response of a Type II closed-loop stable system must *always* have a non-zero overshoot.

4.29 Consider the feedback control system shown in Figure 4.47.

(a) Assume $D_c(s) = K$. What values of K would make the closed-loop system stable? Explain all your reasoning.

(b) Now consider the controller of the form $D_c(s) = \frac{1}{s^n}$ with n being a non-negative integer. For what values of n is the closed-loop system stable? Explain all your reasoning.

Figure 4.47
Unity feedback system



4.30 A feedback control system is shown in Fig. 4.48.

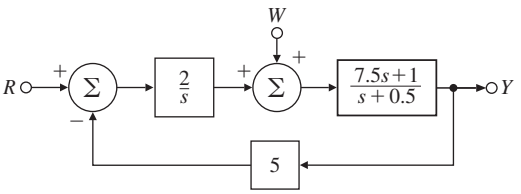
(a) Determine the system type with respect to the reference input.

(b) Compute the steady-state tracking errors, e, for unit step and ramp inputs.

(c) Determine the system type with respect to the disturbance input, w.

(d) Compute the steady-state errors, e, for unit step and ramp disturbance inputs.

Figure 4.48
Feedback system for Problem 4.30



4.31 Consider the closed-loop system shown in Fig. 4.49.

(a) What is the condition on the gain, K, for the closed-loop system to be stable?

(b) What is the system type with respect to the reference input, r?

approximates proportional–integral (PI) control. It generally improves the steady-state error for fixed speed of response by increasing the low-frequency gain and typically degrades stability.

△ • The root locus can be used to analyze successive loop closures by studying two (or more) parameters in succession.

REVIEW QUESTIONS

5.1 Give two definitions for the root locus.

5.2 Define the negative root locus.

5.3 Where are the sections of the (positive) root locus on the real axis?

5.4 What are the angles of departure from two coincident poles at $s = -a$ on the real axis? There are no poles or zeros to the right of $-a$.

5.5 What are the angles of departure from *three* coincident poles at $s = -a$ on the real axis? There are no poles or zeros to the right of $-a$.

5.6 What is the principal effect of a lead compensation on a root locus?

5.7 What is the principal effect of a lag compensation on a root locus in the vicinity of the dominant closed-loop roots?

5.8 What is the principal effect of a lag compensation on the steady-state error to a reference input?

5.9 Why is the angle of departure from a pole near the imaginary axis especially important?

5.10 Define a conditionally stable system.

5.11 Show, with a root-locus argument, that a system having three poles at the origin MUST be either unstable or, at best, conditionally stable.

PROBLEMS

Problems for Section 5.1: Root Locus of a Basic Feedback System

5.1 Set up the listed characteristic equations in the form suited to Evans's root-locus method. Give $L(s)$, $a(s)$, and $b(s)$ and the parameter K in terms of the original parameters in each case. Be sure to select K so $a(s)$ and $b(s)$ are monic in each case, and the degree of $b(s)$ is not greater than that of $a(s)$.

(a) $s + (1/\tau) = 0$ versus parameter τ

(b) $s^2 + cs + c + 1 = 0$ versus parameter c

(c) $(s + c)^3 + A(Ts + 1) = 0$

(i) versus parameter A,

(ii) versus parameter T,

(iii) versus the parameter c, if possible. Say why you can or cannot. Can a plot of the roots be drawn versus c for given constant values of A and T by any means at all?

(d) $1 + \left[k_p + \frac{k_I}{s} + \frac{k_D s}{\tau s+1}\right] G(s) = 0$. Assume $G(s) = A\frac{c(s)}{d(s)}$, where $c(s)$ and $d(s)$ are monic polynomials with the degree of $d(s)$ greater than that of $c(s)$.

(i) versus k_p
(ii) versus k_I
(iii) versus k_D
(iv) versus τ

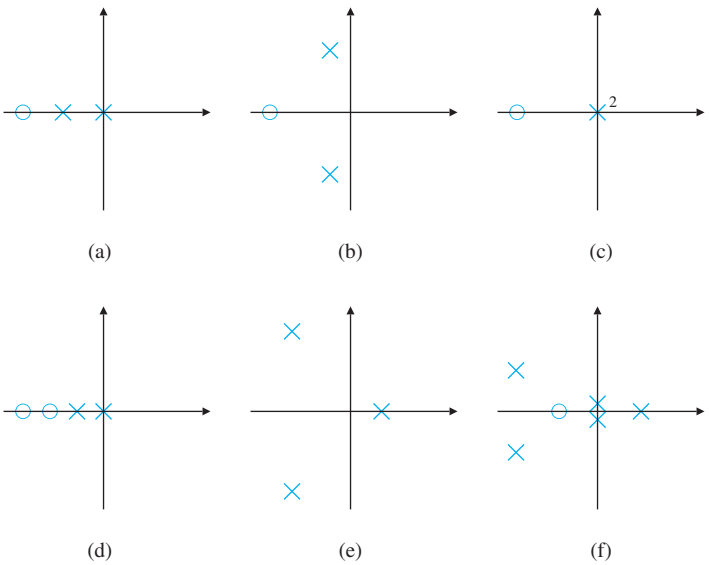
Problems for Section 5.2: Guidelines for Sketching a Root Locus

5.2 Roughly sketch the root loci for the pole–zero maps as shown in Fig. 5.50 without the aid of a computer. Show your estimates of the center and angles of the asymptotes, a rough evaluation of arrival and departure angles for complex poles and zeros, and the loci for positive values of the parameter K. Each pole–zero map is from a characteristic equation of the form

$$1 + K\frac{b(s)}{a(s)} = 0,$$

where the roots of the numerator $b(s)$ are shown as small circles ∘ and the roots of the denominator $a(s)$ are shown as ×'s on the s-plane. Note in Fig. 5.50(c), there are two poles at the origin.

Figure 5.50
Pole–zero maps



5.3 For the characteristic equation

$$1 + \frac{K}{s^2(s+1)(s+5)} = 0,$$

(a) Draw the real-axis segments of the corresponding root locus.
(b) Sketch the asymptotes of the locus for $K \to \infty$.

(f) $L(s) = \frac{(s+1)^2}{s^3(s+4)}$

(g) $L(s) = \frac{(s+1)^2}{s^3(s+10)}$

5.7 *Mixed real and complex poles*. Sketch the root locus with respect to K for the equation $1 + KL(s) = 0$ and the listed choices for $L(s)$. Be sure to give the asymptotes and the arrival and departure angles at any complex zero or pole. After completing each hand sketch, verify your results using Matlab. Turn in your hand sketches and the Matlab results on the same scales.

(a) $L(s) = \frac{(s+3)}{s(s+10)(s^2+2s+2)}$

(b) $L(s) = \frac{(s+3)}{s^2(s+10)(s^2+6s+25)}$

(c) $L(s) = \frac{(s+3)^2}{s^2(s+10)(s^2+6s+25)}$

(d) $L(s) = \frac{(s+3)(s^2+4s+68)}{s^2(s+10)(s^2+4s+85)}$

(e) $L(s) = \frac{[(s+1)^2+1]}{s^2(s+2)(s+3)}$

5.8 *RHP and zeros*. Sketch the root locus with respect to K for the equation $1 + KL(s) = 0$ and the listed choices for $L(s)$. Be sure to give the asymptotes and the arrival and departure angles at any complex zero or pole. After completing each hand sketch, verify your results using Matlab. Turn in your hand sketches and the Matlab results on the same scales.

(a) $L(s) = \frac{s+2}{s+10}\frac{1}{s^2-1}$; the model for a case of magnetic levitation with lead compensation.

(b) $L(s) = \frac{s+2}{s(s+10)}\frac{1}{(s^2-1)}$; the magnetic levitation system with integral control and lead compensation.

(c) $L(s) = \frac{s-1}{s^2}$

(d) $L(s) = \frac{s^2+2s+1}{s(s+20)^2(s^2-2s+2)}$. What is the largest value that can be obtained for the damping ratio of the stable complex roots on this locus?

(e) $L(s) = \frac{(s+2)}{s(s-1)(s+6)^2}$

(f) $L(s) = \frac{1}{(s-1)[(s+2)^2+3]}$

5.9 Put the characteristic equation of the system shown in Fig. 5.51 in root-locus form with respect to the parameter α, and identify the corresponding $L(s)$, $a(s)$, and $b(s)$. When $\alpha = 0.5$, 1.0, and 1.5, find the

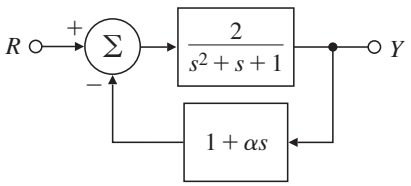


Figure 5.51
Control system for Problem 5.9

closed-loop pole locations, verify your results from the root locus with respect to the parameter α and sketch the corresponding step responses. Use Matlab to check the accuracy of your approximate step responses.

5.10 Use the Matlab function sisotool to study the behavior of the root locus of $1 + KL(s)$ for

$$L(s) = \frac{(s+a)}{s(s+1)(s^2+8s+52)}$$

as the parameter a is varied from 0 to 10, paying particular attention to the region between 2.5 and 3.5. Verify that a multiple root occurs at a complex value of s for some value of a in this range.

5.11 Use Routh's criterion to find the range of the gain K for which the systems in Fig. 5.52 are stable, and use the root locus to confirm your calculations.

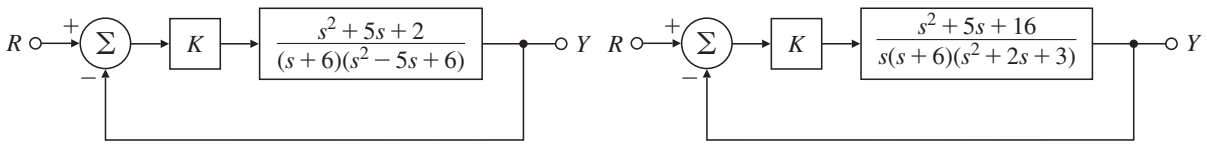


Figure 5.52
Feedback systems for Problem 5.11

5.12 Sketch the root locus for the characteristic equation of the system for which

$$L(s) = \frac{(s+2)}{s^2(s+10)},$$

and determine the value of the root-locus gain for which the complex conjugate poles have the maximum damping ratio. What is the approximate value of the damping?

5.13 For the system in Fig. 5.53,

(a) Find the locus of closed-loop roots with respect to K.

(b) Is there a value of K that will cause all roots to have a damping ratio greater than 0.5?

(c) Find the values of K that yield closed-loop poles with the damping ratio $\zeta = 0.707$.

(d) Use Matlab to plot the response of the resulting design to a reference step.

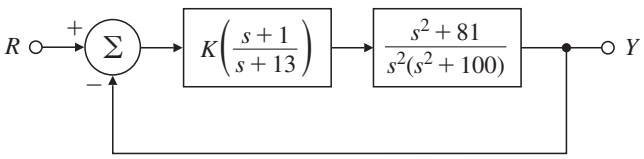


Figure 5.53
Feedback system for Problem 5.13

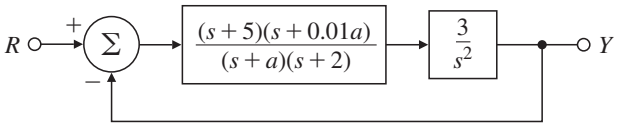


Figure 5.56
Control system for Problem 5.17

5.18 The loop transmission of a system has two poles at $s = -1$ and a zero at $s = -2$. There is a third real-axis pole p located somewhere to the *left* of the zero. Several different root loci are possible, depending on the exact location of the third pole. The extreme cases occur when the pole is located at infinity or when it is located at $s = -2$. Give values for p and sketch the three distinct types of loci.

5.19 For the feedback configuration of Fig. 5.57, use asymptotes, center of asymptotes, angles of departure and arrival, and the Routh array to sketch root loci for the characteristic equations of the listed feedback control systems versus the parameter K. Use Matlab to verify your results.

(a) $G(s) = \dfrac{K}{s(s+2+8j)(s+2-8j)}$, $\quad H(s) = \dfrac{s+1}{s+6}$

(b) $G(s) = \dfrac{K}{s^2}$, $\quad H(s) = \dfrac{s+2}{s+5}$

(c) $G(s) = \dfrac{K(s+4)}{(s+3)}$, $\quad H(s) = \dfrac{s+9}{s+2}$

(d) $G(s) = \dfrac{K(s+2+1j)(s+2-1j)}{s(s+5-7j)(s+5+7j)}$, $\quad H(s) = \dfrac{1}{s+3}$

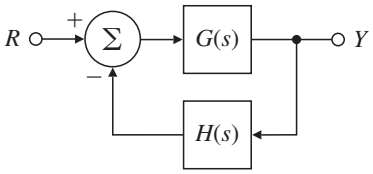


Figure 5.57
Feedback system for Problem 5.19

5.20 Consider the system in Fig. 5.58.

(a) Using Routh's stability criterion, determine all values of K for which the system is stable.

(b) Use Matlab to draw the root locus versus K and find the values of K at the imaginary-axis crossings.

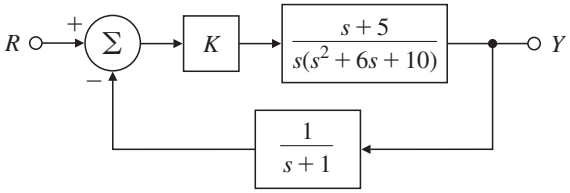


Figure 5.58
Feedback system for Problem 5.20

Problems for Section 5.4: Design Using Dynamic Compensation

5.21 Let

$$G(s) = \frac{1}{s^2 + 7s + 12} \quad \text{and} \quad D_c(s) = K\frac{(s+a)}{s+b}.$$

Using root-locus techniques, find the values for the parameters a, b, and K of the compensation $D_c(s)$ that will produce closed-loop poles at $s = -1.5 \pm 1.5j$ for the system shown in Fig. 5.59.

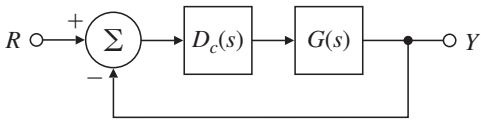


Figure 5.59
Unity feedback system for Problems 5.21–5.27, and 5.32

5.22 Suppose in Fig. 5.59

$$G(s) = \frac{1}{s(s^2 + 3s + 7)} \quad \text{and} \quad D(s) = \frac{K}{s + 3}.$$

Without using Matlab, sketch the root locus with respect to K of the characteristic equation for the closed-loop system, paying particular attention to points that generate multiple roots. Find the value of K at that point, state what the location of the mulitple roots, and how many multiple roots there are.

5.23 Suppose the unity feedback system of Fig. 5.59 has an open-loop plant given by $G(s) = \frac{1}{s(s+1)}$. Design a lead compensation $D_c(s) = K\frac{s+z}{s+p}$ to be added in cascade with the plant so that the dominant poles of the closed-loop system are located at $s = -3.2 \pm 3.2j$.

5.24 Assume that the unity feedback system of Fig. 5.59 has the open-loop plant

$$G(s) = \frac{s + 7}{s(s + 9)(s + 5)}.$$

Design a lag compensation $D_c(s) = K\frac{(s-z)}{s-p}$ to meet the following specifications:

- The step response rise time is to be less than 0.45 sec.
- The step response overshoot is to be less than 5%.
- The steady-state error to a unit ramp input must not exceed 10%.

5.25 A numerically controlled machine tool positioning servomechanism has a normalised and scaled transfer function given by

$$G(s) = \frac{1}{(s + 0.8)(s + 0.5)}.$$

Performance specifications of the system in the unit feedback configuration of Fig. 5.59 are satisfied if the closed-loop poles are located at $s = -1 \pm j2$.

(a) Show that this specification cannot be achieved by choosing proportional control alone, $D_c(s) = k_p$.

(b) Design a lead compensator $D_c(s) = K\frac{s-z}{s-p}$ that will meet the specification.

5.26 A servomechanism position control has the plant transfer function

$$G(s) = \frac{10}{s(s + 1)(s + 10)}.$$

You are to design a series compensation transfer function $D_c(s)$ in the unity feedback configuration to meet the following closed-loop specifications:

- The response to a reference step input is to have no more than 16% overshoot.

5.29 A certain plant with the non-minimum phase transfer function

$$G(s) = \frac{7 - 3s}{s^2 + s + 5},$$

is in a unity positive feedback system with the controller transfer function $D_c(s)$.

(a) Use Matlab to determine a (negative) value for $D_c(s) = K$ so that the closed-loop system with negative feedback has a damping ratio $\zeta = 0.707$.

(b) Use Matlab to plot the system's response to a reference step.

5.30 Consider the rocket-positioning system shown in Fig. 5.61.

(a) Show that if the sensor that measures x has a unity transfer function, the lead compensator

$$H(s) = K\frac{s + 3}{s + 6},$$

stabilizes the system.

(b) Assume that the sensor transfer function is modelled by a single pole with a 0.1 sec time constant and unit DC gain, and $H(s)$ is a PD controller having transfer function $K(s + 1)$. Using the root-locus procedure, find a value for the gain K that will provide the maximum damping ratio while its step response has the lowest settling time.

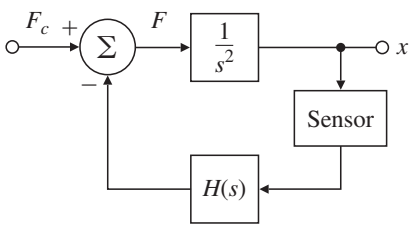


Figure 5.61
Block diagram for rocket-positioning control system

5.31 For the system in Fig. 5.62,

(a) Find the locus of closed-loop roots with respect to K.

(b) Find the maximum value of K for which the system is stable. Assume $K = 0.5$ for the remaining parts of this problem.

(c) What is the steady-state error ($e = r - y$) for a step change in r?

(d) What is the steady-state error in y for a constant disturbance w_1?

(e) What is the steady-state error in y for a constant disturbance w_2?

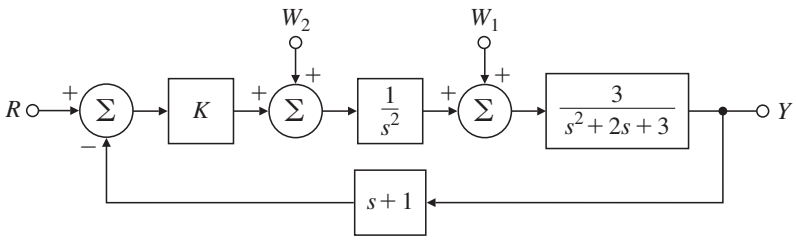


Figure 5.62
Control system for Problem 5.31

(f) If you wished to have more damping, what changes would you make to the system?

5.32 Consider the plant transfer function

$$G(s) = \frac{bs + k}{s^2[mMs^2 + (M + m)bs + (M + m)k]}$$

to be put in the unity feedback loop of Fig. 5.59. This is the transfer function relating the input force $u(t)$ of mass M in the non-collocated sensor and actuator problem. In this problem, we will use root-locus techniques to design a controller $D_c(s)$ so that the closed-loop step response has a rise time of less than 0.5 sec and an overshoot of less than 15%. You may use Matlab for any of the following questions:

(a) Approximate $G(s)$ by assuming that $m \approx 0$, and let $M = 1$, $k = 1$, $b = 0.5$, and $D_c(s) = K$. Can K be chosen to satisfy the performance specifications? Why or why not?

(b) Repeat part (a) assuming $D_c(s) = K(s + z)$, and show that K and z can be chosen to meet the specifications.

(c) Repeat part (b) but with a practical controller given by the transfer function

$$D_c(s) = K\frac{p(s + z)}{s + p}.$$

and using the value for z in part(b), pick p and K so that the step response similar to part (b) is obtained.

(d) Now suppose that the small mass m is not negligible, but is given by $m = M/10$. Check to see if the controller you design in part (c) still meets the given specifications. If not, adjust the controller parameters or suggest a new controller so that the specifications are met.

5.33 Consider the Type 1 system drawn in Fig. 5.63. We would like to design the compensation $D_c(s)$ to meet the following requirements: (1) The steady-state value of y due to a constant unit disturbance w should be less than 0.1, and (2) the damping ratio $\zeta > 0.7$. Using root-locus techniques,

(a) Show that proportional control alone is not adequate.

(b) Show that proportional-derivative control will work.

(c) Find values of the gains k_p and k_D for $D_c(s) = k_p + k_D s$ that meet the design specifications with at least 10% margin.

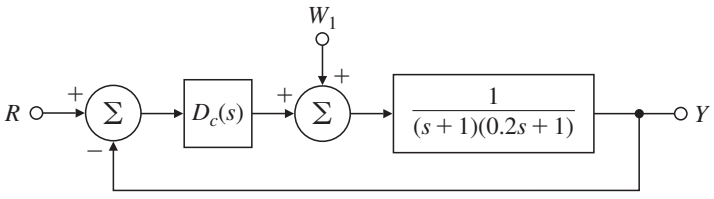


Figure 5.63
Control system for Problem 5.33

△ **5.34** Using a sample rate of 10 Hz, find the $D_c(z)$ that is the discrete equivalent to your $D_c(s)$ from Problem 5.7 using the trapezoid rule. Evaluate the time response using Simulink, and determine whether the damping ratio requirement is met with the digital implementation. (*Note: The material to do this problem is covered in the online Appendix W4.5 at www.pearsonglobaleditions.com or in Chapter 8.*)